

Exercise 3 – Solution Proposal

Electric Double Layer Modeling:
Gouy–Chapman–Stern & Bikerman

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Introduction

In this document, we investigate extensions of the Gouy–Chapman model for the electric double layer. The questions and notation are based on the problem sheet [1]. We focus on three related descriptions:

- the Gouy–Chapman–Stern model, which adds a compact Helmholtz layer,
- the Bikerman model, which introduces finite ion-size effects,
- the Freise model, which allows different cation and anion volume coefficients.

The aim is to compare how these models describe the electric potential, ion concentration profiles, surface free charge density, and double-layer capacitance.

Equations: Surface potential and Debye length

We use the applied surface potential relative to the potential of zero charge,

$$\Delta\phi = \phi_M - \phi_{\text{pzc}} \quad (1)$$

The inverse Debye length is defined as

$$\lambda_D^{-1} = \left(\frac{2e_0^2 N_A c_b}{\varepsilon_0 \varepsilon_S k_B T} \right)^{1/2} \quad (2)$$

Equations: Gouy–Chapman–Stern model

The Gouy–Chapman–Stern model combines a compact Helmholtz layer with a diffuse Gouy–Chapman layer.

In the compact Helmholtz layer,

$$\frac{d^2\phi_H}{dx^2} = 0, \quad 0 \leq x \leq \delta_{\text{HP}}, \quad (3)$$

$$\phi_H(0) = \Delta\phi, \quad (4)$$

$$\phi_H(\delta_{\text{HP}}) = \phi_{\text{HP}}. \quad (5)$$

For a given Helmholtz-plane potential ϕ_{HP} , the Helmholtz-layer potential is

$$\phi_H(x) = \Delta\phi - \frac{x}{\delta_{\text{HP}}} [\Delta\phi - \phi_{\text{HP}}]. \quad (6)$$

In the diffuse layer, $x > \delta_{\text{HP}}$, the Gouy–Chapman solution starts from ϕ_{HP} :

$$\phi_{\text{GC}}(x) = \frac{4k_B T}{e_0} \operatorname{arctanh} \left[\tanh \left(\frac{e_0 \phi_{\text{HP}}}{4k_B T} \right) \exp \left(-\frac{x - \delta_{\text{HP}}}{\lambda_D} \right) \right]. \quad (7)$$

Thus,

$$\phi^{(\text{GCS})}(x) = \begin{cases} \phi_H(x), & 0 \leq x \leq \delta_{\text{HP}}, \\ \phi_{\text{GC}}(x), & x > \delta_{\text{HP}}. \end{cases} \quad (8)$$

Equations: Gouy–Chapman–Stern charge-density matching

The Helmholtz-plane potential ϕ_{HP} is not chosen freely. It is determined by matching the compact-layer and diffuse-layer surface charge densities:

$$\sigma_{\text{H}} = \sigma_{\text{GC}}. \quad (9)$$

That is,

$$\varepsilon_0 \varepsilon_{\text{HP}} \frac{\Delta\phi - \phi_{\text{HP}}}{\delta_{\text{HP}}} = \sqrt{8k_{\text{B}}T\varepsilon_0\varepsilon_{\text{S}}N_{\text{A}}c_{\text{b}}} \sinh\left(\frac{e_0\phi_{\text{HP}}}{2k_{\text{B}}T}\right). \quad (10)$$

The Gouy–Chapman–Stern capacitance can be interpreted as a series combination,

$$\frac{1}{C_{\text{dl}}^{(\text{GCS})}} = \frac{1}{C_{\text{HP}}} + \frac{1}{C_{\text{D}}(\phi_{\text{HP}})}, \quad (11)$$

with

$$C_{\text{HP}} = \frac{\varepsilon_0 \varepsilon_{\text{HP}}}{\delta_{\text{HP}}}, \quad (12)$$

$$C_{\text{D}}(\phi_{\text{HP}}) = \varepsilon_0 \varepsilon_{\text{S}} \lambda_{\text{D}}^{-1} \cosh\left(\frac{e_0\phi_{\text{HP}}}{2k_{\text{B}}T}\right) \quad (13)$$

Equations: Bikerman model

The Bikerman model modifies the Gouy–Chapman model by including finite ion-size effects. Define

$$U(x) = \frac{e_0\phi(x)}{k_{\text{B}}T}, \quad \nu = 2r_0^3 N_{\text{A}} c_{\text{b}} \quad (14)$$

The Bikerman denominator is

$$D_{\text{B}}(U) = 1 + 2\nu \sinh^2\left(\frac{U}{2}\right) \quad (15)$$

The one-dimensional Bikerman equation is

$$\varepsilon_0 \varepsilon_{\text{S}} \frac{d^2\phi}{dx^2} = \frac{2e_0 N_{\text{A}} c_{\text{b}} \sinh\left(\frac{e_0\phi}{k_{\text{B}}T}\right)}{1 + 4r_0^3 N_{\text{A}} c_{\text{b}} \sinh^2\left(\frac{e_0\phi}{2k_{\text{B}}T}\right)}, \quad x \in]0, \infty[, \quad (16)$$

with boundary conditions

$$\phi(0) = \Delta\phi, \quad \lim_{x \rightarrow \infty} \phi(x) = 0 \quad (17)$$

Equations: Bikerman ion concentrations and capacitance

The Bikerman ion concentrations are

$$c_{-}^{(\text{B})}(x) = \frac{c_{\text{b}} \exp(U(x))}{D_{\text{B}}(U(x))} \quad (18)$$

$$c_{+}^{(\text{B})}(x) = \frac{c_{\text{b}} \exp(-U(x))}{D_{\text{B}}(U(x))} \quad (19)$$

The Bikerman surface free charge density can be written as

$$\sigma_{\text{free}}^{(\text{B})}(\Delta\phi) = \text{sgn}(\Delta\phi) \sqrt{\frac{2\varepsilon_0\varepsilon_{\text{S}}k_{\text{B}}T}{r_0^3} \ln \left[1 + 4r_0^3 N_{\text{A}} c_b \sinh^2 \left(\frac{e_0 \Delta\phi}{2k_{\text{B}}T} \right) \right]} \quad (20)$$

The differential capacitance is

$$C_{\text{dl}}^{(\text{B})}(\Delta\phi) = \frac{d\sigma_{\text{free}}^{(\text{B})}}{d\Delta\phi} \quad (21)$$

At $\Delta\phi = 0$, the capacitance is evaluated by the limiting value

$$C_{\text{dl}}^{(\text{B})}(0) = \varepsilon_0\varepsilon_{\text{S}}\lambda_D^{-1} \quad (22)$$

Equations: Freise model

The Freise model extends the Bikerman model by allowing different cation and anion volume factors. Define

$$U(x) = \frac{e_0\phi(x)}{k_{\text{B}}T}, \quad \alpha = r_0^3 N_{\text{A}} c_b \quad (23)$$

The Freise denominator is

$$D_{\text{F}}(U) = 1 + \alpha [\gamma_{\text{c}} (e^{-U} - 1) + \gamma_{\text{a}} (e^U - 1)] \quad (24)$$

The one-dimensional Freise equation is

$$\varepsilon_0\varepsilon_{\text{S}} \frac{d^2\phi}{dx^2} = \frac{2e_0 N_{\text{A}} c_b \sinh \left(\frac{e_0\phi}{k_{\text{B}}T} \right)}{1 + r_0^3 N_{\text{A}} c_b [\gamma_{\text{c}} (e^{-e_0\phi/k_{\text{B}}T} - 1) + \gamma_{\text{a}} (e^{+e_0\phi/k_{\text{B}}T} - 1)]}, \quad x \in]0, \infty[\quad (25)$$

with boundary conditions

$$\phi(0) = \Delta\phi, \quad \lim_{x \rightarrow \infty} \phi(x) = 0 \quad (26)$$

Equations: Freise ion concentrations and capacitance

The Freise ion concentrations are

$$c_+^{(\text{F})}(x) = \frac{c_b e^{-U(x)}}{D_{\text{F}}(U(x))} \quad (27)$$

$$c_-^{(\text{F})}(x) = \frac{c_b e^{+U(x)}}{D_{\text{F}}(U(x))} \quad (28)$$

For $\gamma_{\text{c}} = \gamma_{\text{a}} = 1$, the symmetric Bikerman model is recovered.

The surface free charge density is defined by

$$\sigma_{\text{free}} = -\varepsilon_0\varepsilon_{\text{S}} \left. \frac{d\phi}{dx} \right|_{x=0^+} \quad (29)$$

The differential capacitance is

$$C_{\text{dl}}^{(\text{F})} = \frac{d\sigma_{\text{free}}^{(\text{F})}}{d\Delta\phi} \quad (30)$$

If $\gamma_c \neq \gamma_a$, the surface charge density and capacitance are generally asymmetric around the potential of zero charge.

Table 1: Fundamental constants

Symbol	Value	Unit
N_A	6.022×10^{23}	mol^{-1}
k_B	1.381×10^{-23}	J K^{-1}
e_0	1.602×10^{-19}	C
ε_0	8.854×10^{-12}	F m^{-1}

* Values are given in SI units.

Table 2: Model parameters and variables

Symbol	Unit	Meaning
$\phi_M - \phi_{\text{pzc}}$	V	Applied metal potential relative to the potential of zero charge.
T	K	Temperature.
ε_S	—	Relative permittivity of the solution region.
ε_{HP}	—	Relative permittivity of the compact Helmholtz layer.
c_b	mol m^{-3}	Bulk ion concentration.
λ_D	m	Debye length.
δ_{HP}	m	Helmholtz-plane distance.
r_0	m	Effective ion or lattice size parameter.
γ_c	—	Cation volume factor.
γ_a	—	Anion volume factor.
x	m	Distance from the metal surface into the electrolyte.

1 Model setup: Gouy–Chapman–Stern

Question: 1 Gouy–Chapman–Stern model setup

For the Gouy–Chapman–Stern model, please answer the following questions:

- 1.1) How does the electric potential change when varying the Stern thickness δ_{HP} ?
- 1.2) How does increasing δ_{HP} change the potential drop inside the compact layer?
- 1.3) How do the cation and anion concentrations change when the Stern layer is introduced?
- 1.4) Why must the compact-layer and diffuse-layer surface charge densities be equal?
- 1.5) How is ϕ_{HP} determined from this charge-continuity condition?
- 1.6) Why does the GCS capacitance behave like capacitors in series? What would the capacitance be if we added another layer?
- 1.7) How well does the Gouy–Chapman–Stern capacitance reproduce the experimental Valette data, and where does it start to deviate?

Solution: 1 Gouy–Chapman–Stern model setup

For convenience, we introduce the applied surface potential relative to the potential of zero charge,

$$\Delta\phi = \phi_{\text{M}} - \phi_{\text{pzc}} \quad (31)$$

The Gouy–Chapman–Stern model separates the double layer into a compact Helmholtz layer and a diffuse Gouy–Chapman layer. The potential at the boundary between these two regions is denoted by ϕ_{HP} .

1.1) Electric potential as a function of δ_{HP}

The compact layer occupies the interval

$$0 \leq x \leq \delta_{\text{HP}} \quad (32)$$

In this region, no mobile ions are present and the potential is linear:

$$\phi_{\text{H}}(x) = \Delta\phi - \frac{x}{\delta_{\text{HP}}} [\Delta\phi - \phi_{\text{HP}}] \quad (33)$$

The diffuse Gouy–Chapman layer starts at the Helmholtz-plane potential ϕ_{HP} :

$$\phi_{\text{GC}}(\delta_{\text{HP}}) = \phi_{\text{HP}}, \quad \lim_{x \rightarrow \infty} \phi_{\text{GC}}(x) = 0 \quad (34)$$

Thus, increasing δ_{HP} changes how much of the total potential drop is taken up in the compact layer. For fixed applied potential $\Delta\phi$, a thicker compact layer decreases the compact-layer capacitance,

$$C_{\text{HP}} = \frac{\varepsilon_0 \varepsilon_{\text{HP}}}{\delta_{\text{HP}}} \quad (35)$$

Therefore, the compact layer becomes less capacitive as δ_{HP} increases.

As a result, more of the applied potential is dropped across the compact layer and the Helmholtz-plane potential ϕ_{HP} generally becomes smaller in magnitude. Since the diffuse layer starts from ϕ_{HP} , the diffuse potential and the near-surface ion accumulation in the solution are reduced.

Hence,

$$\delta_{\text{HP}} \uparrow \implies C_{\text{HP}} \downarrow \implies |\phi_{\text{HP}}| \text{ decreases for fixed } \Delta\phi$$

This behavior is visible in Figure 1, where increasing δ_{HP} changes the decay of the potential away from the metal surface.

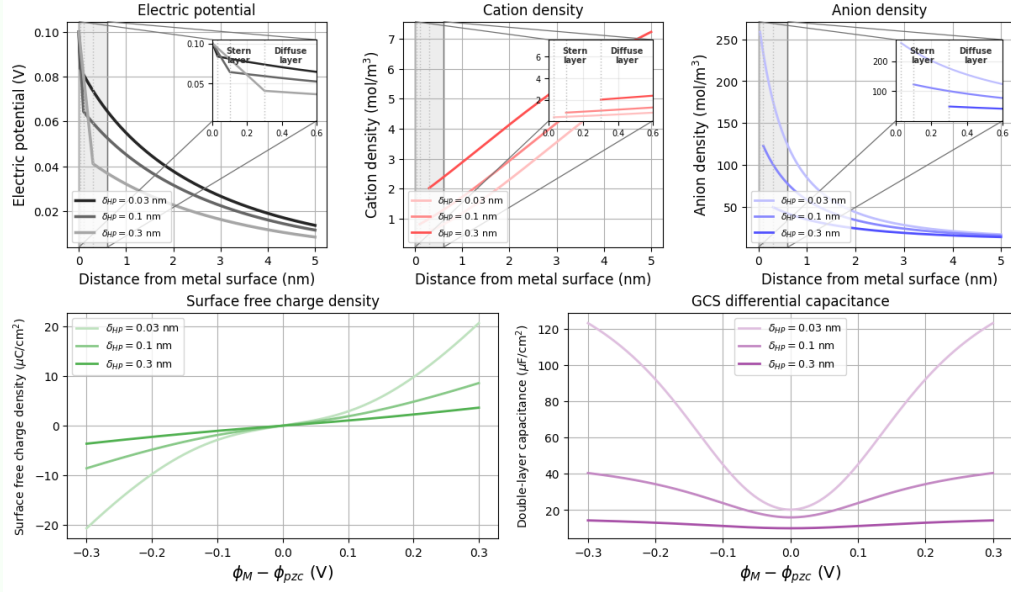


Figure 1: Numerical Gouy–Chapman–Stern profiles for different Stern-layer thicknesses. The compact layer changes the Helmholtz-plane potential and therefore the diffuse-layer potential and ion concentration profiles.

1.2) Potential drop split

The total potential drop is split into a compact Helmholtz contribution and a diffuse Gouy–Chapman contribution:

$$\Delta\phi = \underbrace{[\Delta\phi - \phi_{\text{HP}}]}_{\text{Helmholtz drop}} + \underbrace{[\phi_{\text{HP}} - 0]}_{\text{diffuse Gouy–Chapman drop}} \quad (36)$$

Equivalently,

$$\phi_{\text{M}} - \phi_{\text{pzc}} = [(\phi_{\text{M}} - \phi_{\text{pzc}}) - \phi_{\text{HP}}] + \phi_{\text{HP}} \quad (37)$$

The first term is the potential drop across the compact Helmholtz layer. The second term is the potential drop across the diffuse layer, from ϕ_{HP} to the bulk value zero. The corresponding redistribution of the potential drop between the compact and diffuse regions is illustrated in Figure 1.

1.3) Equality of compact-layer and diffuse-layer charge densities

The compact Helmholtz layer and the diffuse Gouy–Chapman layer meet at the Helmholtz plane. The same surface charge must be supported by both descriptions of the interface.

Therefore, the charge density obtained from the compact-layer electric field must be equal to the charge density obtained from the diffuse-layer Gouy–Chapman solution:

$$\sigma_H = \sigma_{GC} \quad (38)$$

The Helmholtz contribution is

$$\sigma_H = \varepsilon_0 \varepsilon_{HP} \frac{\Delta\phi - \phi_{HP}}{\delta_{HP}} \quad (39)$$

The diffuse-layer contribution is

$$\sigma_{GC} = \sqrt{8k_B T \varepsilon_0 \varepsilon_S N_A c_b} \sinh\left(\frac{e_0 \phi_{HP}}{2k_B T}\right) \quad (40)$$

Thus,

$$\boxed{\sigma_H = \sigma_{GC}}$$

is the coupling condition between the two parts of the GCS model. The concentration panels in Figure 1 show that changing the Stern layer modifies the diffuse-layer potential and therefore the near-surface cation and anion densities.

1.4) Determination of ϕ_{HP}

The Helmholtz-plane potential ϕ_{HP} is not an independent input. It is determined by the charge-density matching condition:

$$\varepsilon_0 \varepsilon_{HP} \frac{\Delta\phi - \phi_{HP}}{\delta_{HP}} = \sqrt{8k_B T \varepsilon_0 \varepsilon_S N_A c_b} \sinh\left(\frac{e_0 \phi_{HP}}{2k_B T}\right) \quad (41)$$

For a given applied potential $\Delta\phi$, this equation is solved for ϕ_{HP} . Once ϕ_{HP} is known, the complete potential profile is given by

$$\phi^{(GCS)}(x) = \begin{cases} \phi_H(x), & 0 \leq x \leq \delta_{HP} \\ \phi_{GC}(x), & x > \delta_{HP} \end{cases} \quad (42)$$

Therefore,

$$\boxed{\phi_{HP} \text{ is fixed by charge-density continuity, not chosen freely.}}$$

1.5) Surface charge density at the potential of zero charge

At the potential of zero charge,

$$\phi_M = \phi_{pzc} \quad (43)$$

and therefore

$$\Delta\phi = \phi_M - \phi_{pzc} = 0 \quad (44)$$

The matching condition becomes

$$\varepsilon_0 \varepsilon_{HP} \frac{0 - \phi_{HP}}{\delta_{HP}} = \sqrt{8k_B T \varepsilon_0 \varepsilon_S N_A c_b} \sinh\left(\frac{e_0 \phi_{HP}}{2k_B T}\right) \quad (45)$$

The solution is

$$\phi_{\text{HP}} = 0. \quad (46)$$

Inserting this into the Helmholtz charge density gives

$$\sigma_{\text{H}} = \varepsilon_0 \varepsilon_{\text{HP}} \frac{0 - 0}{\delta_{\text{HP}}} = 0 \quad (47)$$

Similarly, the diffuse-layer expression gives

$$\sigma_{\text{GC}} = \sqrt{8k_{\text{B}}T\varepsilon_0\varepsilon_{\text{S}}N_{\text{A}}c_{\text{b}}} \sinh(0) = 0 \quad (48)$$

Hence,

$$\boxed{\phi_{\text{M}} = \phi_{\text{pzc}} \implies \sigma_{\text{free}}^{(\text{GCS})} = 0}$$

1.6) Derivation of the generalized series capacitance

For the Gouy–Chapman–Stern model, the total potential drop is split into one compact Helmholtz contribution and one diffuse Gouy–Chapman contribution:

$$\Delta\phi = \Delta\phi_{\text{H}} + \Delta\phi_{\text{D}}, \quad (49)$$

with

$$\Delta\phi_{\text{H}} = \Delta\phi - \phi_{\text{HP}} \quad (50\text{a})$$

$$\Delta\phi_{\text{D}} = \phi_{\text{HP}} \quad (50\text{b})$$

For an arbitrary choice of ϕ_{HP} , the compact-layer charge density and diffuse-layer charge density are not necessarily equal. The correct Helmholtz-plane potential is found by imposing the matching condition

$$\sigma_{\text{H}}(\Delta\phi - \phi_{\text{HP}}) = \sigma_{\text{D}}(\phi_{\text{HP}}) \quad (51)$$

Only after this condition has been satisfied do both regions correspond to the same surface free charge density:

$$\sigma_{\text{H}} = \sigma_{\text{D}} = \sigma_{\text{free}} \quad (52)$$

Now consider a small variation around such a consistent solution. Since both layers then describe the same interfacial charge, the differential change in the total potential drop is

$$\text{d}\Delta\phi = \text{d}\Delta\phi_{\text{H}} + \text{d}\Delta\phi_{\text{D}} \quad (53)$$

The local differential capacitance of each contribution is defined by

$$C_i = \frac{\text{d}\sigma_{\text{free}}}{\text{d}\Delta\phi_i} \quad (54)$$

This is a local definition around the matched solution. It does not require a global inverse of the charge–potential curve, but it does require a finite and nonzero local slope. Therefore,

$$\text{d}\Delta\phi_i = \frac{1}{C_i} \text{d}\sigma_{\text{free}} \quad (55)$$

For the two-contribution GCS case, this gives

$$d\Delta\phi = \frac{1}{C_{\text{HP}}}d\sigma_{\text{free}} + \frac{1}{C_{\text{D}}(\phi_{\text{HP}})}d\sigma_{\text{free}} \quad (56)$$

Hence,

$$d\Delta\phi = \left[\frac{1}{C_{\text{HP}}} + \frac{1}{C_{\text{D}}(\phi_{\text{HP}})} \right] d\sigma_{\text{free}} \quad (57)$$

The total differential capacitance is

$$C_{\text{dl}}^{(\text{GCS})} = \frac{d\sigma_{\text{free}}}{d\Delta\phi} \quad (58)$$

Thus,

$$\boxed{\frac{1}{C_{\text{dl}}^{(\text{GCS})}} = \frac{1}{C_{\text{HP}}} + \frac{1}{C_{\text{D}}(\phi_{\text{HP}})}}$$

Here,

$$C_{\text{HP}} = \frac{\varepsilon_0 \varepsilon_{\text{HP}}}{\delta_{\text{HP}}} \quad (59a)$$

$$C_{\text{D}}(\phi_{\text{HP}}) = \frac{d\sigma_{\text{D}}(\phi_{\text{HP}})}{d\phi_{\text{HP}}} = \varepsilon_0 \varepsilon_{\text{S}} \lambda_{\text{D}}^{-1} \cosh\left(\frac{e_0 \phi_{\text{HP}}}{2k_{\text{B}}T}\right) \quad (59b)$$

General case. Now suppose the interface contains N compact layers in series with one diffuse layer. The total potential drop is

$$\Delta\phi = \sum_{i=1}^N \Delta\phi_i + \Delta\phi_{\text{D}} \quad (60)$$

For arbitrary internal potentials, the charge densities of the layers are not automatically equal. The correct internal potentials are those for which all charge-density matching conditions are satisfied. At such a consistent solution,

$$\sigma_1 = \sigma_2 = \dots = \sigma_N = \sigma_{\text{D}} = \sigma_{\text{free}} \quad (61)$$

Taking a small variation around this matched solution gives

$$d\Delta\phi = \sum_{i=1}^N d\Delta\phi_i + d\Delta\phi_{\text{D}}. \quad (62)$$

Using the local capacitance definition

$$d\Delta\phi_i = \frac{1}{C_i}d\sigma_{\text{free}}, \quad d\Delta\phi_{\text{D}} = \frac{1}{C_{\text{D}}}d\sigma_{\text{free}} \quad (63)$$

we obtain

$$d\Delta\phi = \left[\frac{1}{C_{\text{D}}} + \sum_{i=1}^N \frac{1}{C_i} \right] d\sigma_{\text{free}} \quad (64)$$

Therefore, the generalized series capacitance is

$$\frac{1}{C_{\text{dl}}} = \frac{1}{C_{\text{D}}} + \sum_{i=1}^N \frac{1}{C_i}$$

Equivalently,

$$C_{\text{dl}} = \left[\frac{1}{C_{\text{D}}} + \sum_{i=1}^N \frac{1}{C_i} \right]^{-1}$$

The ordinary Gouy–Chapman–Stern model corresponds to $N = 1$:

$$\frac{1}{C_{\text{dl}}^{(\text{GCS})}} = \frac{1}{C_{\text{HP}}} + \frac{1}{C_{\text{D}}^{(\text{GC})}} \quad (65)$$

If one added another compact layer, then $N = 2$. Therefore,

$$\frac{1}{C_{\text{dl}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_{\text{D}}} \quad (66)$$

This directly answers the question of what the capacitance would be after adding one additional compact layer^a.

1.7) Comparison with experimental Valette data

The Gouy–Chapman–Stern model improves upon the pure Gouy–Chapman model by adding a compact Helmholtz capacitance. Since the compact layer appears in series with the diffuse layer,

$$\frac{1}{C_{\text{dl}}^{(\text{GCS})}} = \frac{1}{C_{\text{HP}}} + \frac{1}{C_{\text{D}}(\phi_{\text{HP}})} \quad (67)$$

Therefore, if both capacitances are finite,

$$C_{\text{dl}}^{(\text{GCS})} < C_{\text{D}} \quad (68)$$

Furthermore, at large applied potentials, the diffuse-layer capacitance grows strongly:

$$C_{\text{D}}(\phi_{\text{HP}}) = \varepsilon_0 \varepsilon_{\text{S}} \lambda_{\text{D}}^{-1} \cosh \left(\frac{e_0 \phi_{\text{HP}}}{2k_{\text{B}} T} \right) \quad (69)$$

Thus, for large $|\phi_{\text{M}} - \phi_{\text{pzc}}|$, one typically has

$$C_{\text{D}}(\phi_{\text{HP}}) \gg C_{\text{HP}} \quad (70)$$

In this limit,

$$\frac{1}{C_{\text{D}}(\phi_{\text{HP}})} \rightarrow 0 \quad (71)$$

and therefore

$$C_{\text{dl}}^{(\text{GCS})} \rightarrow C_{\text{HP}} \quad (72)$$

Hence, the compact Helmholtz layer controls the high-potential limiting capacitance of the GCS model.

Compared with experimental Valette capacitance data, the GCS model can improve the overall capacitance scale because the compact layer prevents the capacitance from growing as strongly as in the pure Gouy–Chapman model. In particular, the convergence toward C_{HP} at large $|\phi_{\text{M}} - \phi_{\text{pzc}}|$ gives a much flatter high-potential behavior than the pure diffuse-layer prediction.

However, the agreement is still limited. The GCS model still uses a continuum dielectric description and a Gouy–Chapman diffuse layer. It does not include finite ion size, specific adsorption, solvent structure, dielectric saturation, or ion–ion correlations. Therefore, deviations from the Valette data are expected, especially in regions where real interfacial structure dominates the measured capacitance.

Thus,

$$\text{GCS improves the capacitance scale and gives } C_{\text{dl}}^{(\text{GCS})} \rightarrow C_{\text{HP}} \text{ at large } |\Delta\phi|$$

but

$$\text{it is still incomplete for a quantitative description of real interfaces}$$

As shown in Figure 2, the GCS model can improve the overall capacitance scale because the compact layer prevents the capacitance from growing as strongly as in the pure Gouy–Chapman model.

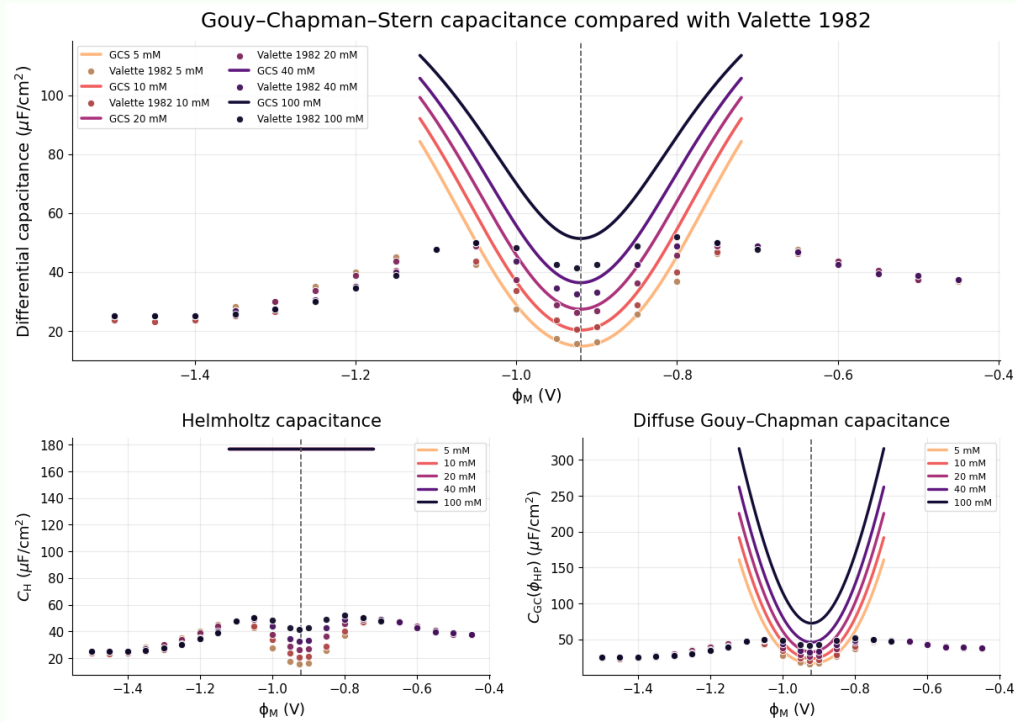


Figure 2: Comparison between Gouy–Chapman–Stern capacitance and experimental Valette data. At large applied potentials, the GCS capacitance approaches the compact-layer capacitance C_{HP} , which improves the high-potential behavior compared with pure Gouy–Chapman. Remaining deviations indicate missing interfacial physics.

^aA note from Octavius: Deriving the 2-compact layer directly would also answer the question. I added the general derivation, since the general series capacitance derivation only differs by a finite sum.

2 Model setup: Bikerman

Question: 2 Bikerman model setup

For the Bikerman model, please answer the following questions:

- 2.1) How does the electric potential change when varying the ion-size parameter r_0 ?
- 2.2) How does the Bikerman model differ from Gouy–Chapman near the metal surface?
- 2.3) How do the cation and anion concentrations change when r_0 increases?
- 2.4) Why do the concentrations no longer diverge at large applied potentials?
- 2.5) How does the finite-size correction affect the surface free charge density and the differential capacitance?
- 2.6) How well does the Bikerman capacitance reproduce the experimental Valette data?

Solution: 2 Bikerman model setup

For convenience, we again introduce

$$\Delta\phi = \phi_M - \phi_{pzc}, \quad U(x) = \frac{e_0\phi(x)}{k_B T} \quad (73)$$

The Bikerman model modifies the Gouy–Chapman model by introducing a finite ion-size parameter r_0 . The corresponding steric denominator is

$$D_B(U) = 1 + 2\nu \sinh^2\left(\frac{U}{2}\right), \quad \nu = 2r_0^3 N_A c_b \quad (74)$$

For $r_0 \rightarrow 0$, one obtains $\nu \rightarrow 0$ and therefore

$$D_B(U) \rightarrow 1 \quad (75)$$

In this limit, the Gouy–Chapman model is recovered.

2.1) Electric potential as a function of r_0

The Bikerman equation is

$$\varepsilon_0 \varepsilon_S \frac{d^2\phi}{dx^2} = \frac{2e_0 N_A c_b \sinh\left(\frac{e_0\phi}{k_B T}\right)}{1 + 4r_0^3 N_A c_b \sinh^2\left(\frac{e_0\phi}{2k_B T}\right)} \quad (76)$$

The boundary conditions are

$$\phi(0) = \Delta\phi, \quad \lim_{x \rightarrow \infty} \phi(x) = 0 \quad (77)$$

Thus, changing r_0 does not change the imposed surface potential $\phi(0) = \Delta\phi$. Instead, it changes how the potential decays away from the surface.

For larger r_0 , the finite-size denominator becomes larger at high $|U|$. This reduces the local charge density that can screen the surface. Consequently, the screening becomes less concentrated directly at the surface and the potential typically decays more slowly into the electrolyte.

Hence,

$$r_0 \uparrow \implies \text{stronger finite-size limitation and broader screening profile}$$

This behavior is visible in Figure 3, where changing r_0 modifies the potential decay away from the metal surface.

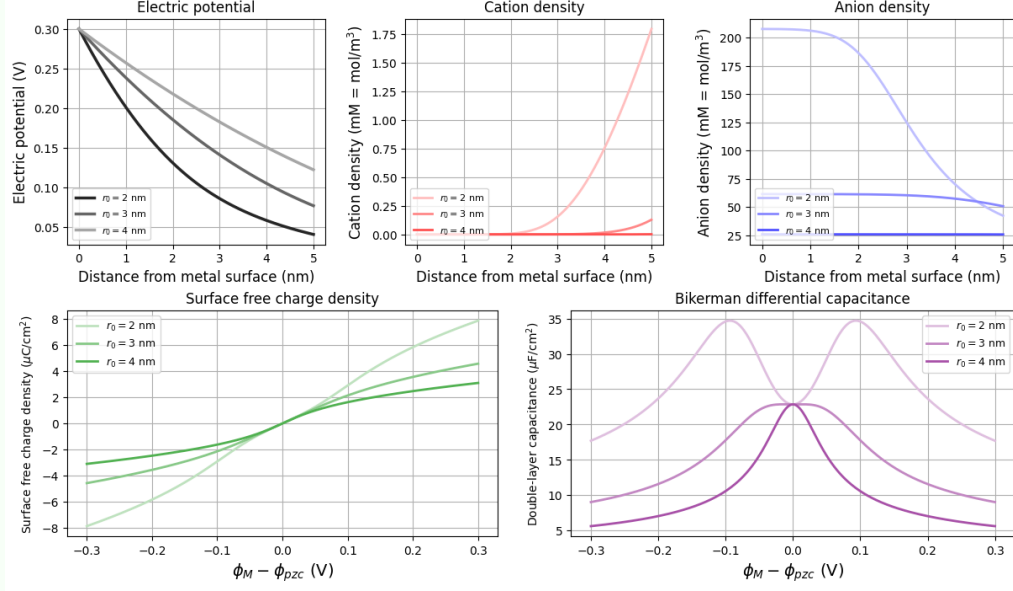


Figure 3: Numerical Bikerman profiles for different ion-size parameters r_0 . Increasing r_0 limits the near-surface ion concentration and changes the potential screening profile.

2.2) Difference from Gouy–Chapman near the metal surface

In the Gouy–Chapman model, ions are treated as point-like particles. The concentrations are simple Boltzmann factors,

$$c_-(x) = c_b e^{U(x)} \quad (78)$$

$$c_+(x) = c_b e^{-U(x)} \quad (79)$$

Therefore, for large $|U|$, the counter-ion concentration can become very large.

In the Bikerman model, the concentrations are divided by the steric denominator:

$$c_-^{(B)}(x) = \frac{c_b e^{U(x)}}{D_B(U(x))} \quad (80)$$

$$c_+^{(B)}(x) = \frac{c_b e^{-U(x)}}{D_B(U(x))} \quad (81)$$

Thus, close to a highly charged metal surface, the Bikerman model suppresses the unphysical counter-ion accumulation predicted by Gouy–Chapman.

The main difference is therefore

$$\text{Gouy–Chapman has point ions, whereas Bikerman includes finite ion size}$$

2.3) Concentrations as r_0 increases

For a positive surface potential,

$$\Delta\phi > 0 \quad (82)$$

the electric potential is positive close to the surface. Therefore, anions are the counter-ions and cations are the co-ions:

$$c_-^{(B)}(x) = \frac{c_b e^{U(x)}}{D_B(U(x))}, \quad c_+^{(B)}(x) = \frac{c_b e^{-U(x)}}{D_B(U(x))} \quad (83)$$

Increasing r_0 increases ν and therefore increases the denominator $D_B(U)$ for large $|U|$. Hence, the near-surface anion accumulation is reduced compared with the Gouy–Chapman case.

For a negative surface potential,

$$\Delta\phi < 0 \quad (84)$$

the roles are reversed. Cations are the counter-ions and anions are the co-ions. Increasing r_0 then reduces the near-surface cation accumulation.

In summary,

$r_0 \uparrow \implies$ counter-ion peaks become smaller and less singular
--

The ion profiles become less extreme near the metal surface. The concentration panels in Figure 3 show that increasing r_0 reduces the near-surface counter-ion peak and makes the ion profiles less extreme.

2.4) Absence of concentration divergence

The reason that the Bikerman concentrations no longer diverge is first of all mathematical: the denominator grows with the same exponential order as the counter-ion Boltzmann factor.

For a large positive dimensionless potential,

$$U \rightarrow +\infty \quad (85)$$

the anion is the counter-ion and

$$c_-^{(B)} = \frac{c_b e^U}{1 + 2\nu \sinh^2\left(\frac{U}{2}\right)} \quad (86)$$

Using

$$\sinh^2\left(\frac{U}{2}\right) \sim \frac{1}{4}e^U \quad \text{for } U \rightarrow +\infty \quad (87)$$

we obtain

$$D_B(U) = 1 + 2\nu \sinh^2\left(\frac{U}{2}\right) \sim \frac{\nu}{2}e^U \quad (88)$$

Therefore,

$$c_-^{(B)} \sim \frac{c_b e^U}{\frac{\nu}{2}e^U} = \frac{2c_b}{\nu} \quad (89)$$

Since

$$\nu = 2r_0^3 N_A c_b \quad (90)$$

this gives

$$c_-^{(B)} \rightarrow \frac{1}{r_0^3 N_A} \quad (91)$$

Similarly, for a large negative dimensionless potential,

$$U \rightarrow -\infty \quad (92)$$

the cation is the counter-ion and one obtains

$$c_+^{(B)} \rightarrow \frac{1}{r_0^3 N_A} \quad (93)$$

Thus, mathematically,

$$c_{\pm}^{(B)} \text{ approaches a finite limiting value instead of diverging}$$

Physically, this finite limiting value corresponds to a finite-volume or crowding effect. The ions are no longer treated as point particles. Instead, each ion occupies an effective volume of order r_0^3 . Therefore, the concentration cannot increase without bound near the metal surface.

The maximum concentration scale is

$$c_{\max} = \frac{1}{r_0^3 N_A} \quad (94)$$

This is the concentration corresponding to filling the available volume with ions of effective size r_0^3 .

Hence,

$$\text{Mathematically, the denominator cancels the exponential divergence}$$

and

$$\text{Physically, this represents finite ion volume and crowding near the surface}$$

2.5) Effect on surface charge density and capacitance

The surface free charge density is defined by

$$\sigma_{\text{free}} = -\varepsilon_0 \varepsilon_S \left. \frac{d\phi}{dx} \right|_{x=0^+} \quad (95)$$

For the Bikerman model, the surface free charge density can be written as

$$\sigma_{\text{free}}^{(B)}(\Delta\phi) = \text{sgn}(\Delta\phi) \sqrt{\frac{2\varepsilon_0 \varepsilon_S k_B T}{r_0^3} \ln \left[1 + 4r_0^3 N_A c_b \sinh^2 \left(\frac{e_0 \Delta\phi}{2k_B T} \right) \right]} \quad (96)$$

At the potential of zero charge,

$$\Delta\phi = 0 \quad (97)$$

Using

$$\sinh(0) = 0 \quad (98)$$

we obtain

$$\sigma_{\text{free}}^{(\text{B})}(0) = 0 \quad (99)$$

Therefore,

$$\boxed{\phi_{\text{M}} = \phi_{\text{pzc}} \implies \sigma_{\text{free}}^{(\text{B})} = 0}$$

The differential capacitance is

$$C_{\text{dl}}^{(\text{B})}(\Delta\phi) = \frac{d\sigma_{\text{free}}^{(\text{B})}}{d\Delta\phi} \quad (100)$$

At $\Delta\phi = 0$, the derivative of the expression above should be evaluated by its limiting value. This gives

$$C_{\text{dl}}^{(\text{B})}(0) = \varepsilon_0 \varepsilon_{\text{S}} \lambda_D^{-1} \quad (101)$$

Compared with Gouy–Chapman, the Bikerman surface charge density grows more slowly at large $|\Delta\phi|$. Consequently, the Bikerman capacitance is reduced at large applied potentials. Physically, this happens because finite ion size limits the amount of charge that can be packed close to the surface.

Thus,

$$\boxed{\text{finite ion size reduces the high-voltage growth of } \sigma_{\text{free}} \text{ and } C_{\text{dl}}}$$

The corresponding changes in σ_{free} and $C_{\text{dl}}^{(\text{B})}$ are also shown in Figure 3.

2.6) Comparison with experimental Valette data

The Bikerman model improves upon the Gouy–Chapman model by preventing unphysical near-surface ion accumulation. Therefore, it can reduce the overly strong increase in capacitance predicted by point-ion Gouy–Chapman theory at large applied potentials. In comparison with the experimental Valette capacitance data, this finite-size correction can improve the qualitative behavior away from the potential of zero charge. However, the Bikerman model is still limited. It includes finite ion size, but it does not include a compact Stern layer, specific adsorption, dielectric saturation, solvent structure, or ion–ion correlations.

Thus,

$$\boxed{\text{Bikerman improves the high-potential behavior by limiting ion crowding}}$$

but

$$\boxed{\text{it is still not a complete quantitative model for real interfaces}}$$

As shown in Figure 4, the Bikerman model improves upon the Gouy–Chapman model by preventing unphysical near-surface ion accumulation. Therefore, it can reduce the overly strong increase in capacitance predicted by point-ion Gouy–Chapman theory at large applied potentials.

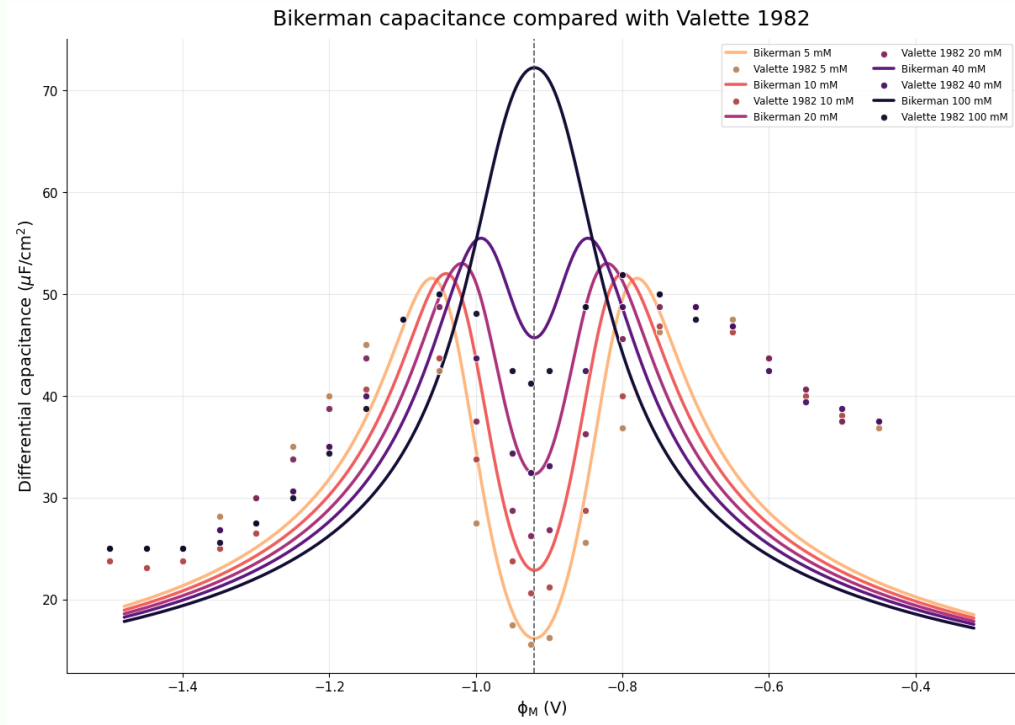


Figure 4: Comparison between Bikerman capacitance and experimental Valette data. The finite-size correction limits the high-potential capacitance growth, but additional interfacial effects are still needed for quantitative agreement.

3 Model setup: Freise

Question: 3 Different size coefficients – Freise model

Optional / extra question. The Freise model extends the Bikerman model by allowing different cation and anion volume factors.

- 3.1) What changes when $\gamma_c \neq \gamma_a$?
- 3.2) How does increasing γ_a affect anions near a positively charged surface?
- 3.3) How does increasing γ_c affect cations near a negatively charged surface?
- 3.4) Why does σ_{free} become asymmetric with respect to $\phi_M - \phi_{\text{pzc}} = 0$?
- 3.5) Why does C_{dl} become asymmetric when $\gamma_c \neq \gamma_a$?
- 3.6) Does Freise improve agreement with Valette data compared with symmetric Bikerman?

Solution: 3 Different size coefficients – Freise model

The Freise model extends the Bikerman model by allowing the cation and anion to contribute differently to the steric denominator. We define

$$\Delta\phi = \phi_M - \phi_{\text{pzc}}, \quad U(x) = \frac{e_0\phi(x)}{k_B T}, \quad \alpha = r_0^3 N_A c_b \quad (102)$$

The Freise denominator is

$$D_F(U) = 1 + \alpha [\gamma_c (e^{-U} - 1) + \gamma_a (e^U - 1)] \quad (103)$$

The ion concentrations are

$$c_+^{(F)}(x) = \frac{c_b e^{-U(x)}}{D_F(U(x))} \quad (104)$$

$$c_-^{(F)}(x) = \frac{c_b e^{+U(x)}}{D_F(U(x))} \quad (105)$$

For

$$\gamma_c = \gamma_a = 1 \quad (106)$$

the symmetric Bikerman model is recovered.

3.1) What changes when $\gamma_c \neq \gamma_a$?

If

$$\gamma_c \neq \gamma_a \quad (107)$$

then the cation and anion no longer have the same effective volume contribution. The steric denominator is no longer symmetric with respect to changing

$$U \longrightarrow -U \quad (108)$$

Indeed,

$$D_F(U) = 1 + \alpha [\gamma_c (e^{-U} - 1) + \gamma_a (e^U - 1)] \quad (109)$$

whereas

$$D_F(-U) = 1 + \alpha [\gamma_c (e^U - 1) + \gamma_a (e^{-U} - 1)] \quad (110)$$

These two expressions are equal only if $\gamma_c = \gamma_a$.

Therefore, the model no longer treats positive and negative applied potentials as mirror images. The positive branch and the negative branch of the double layer can behave differently.

Hence,

$$\gamma_c \neq \gamma_a \implies \text{asymmetric concentration, charge, and capacitance profiles}$$

This asymmetry is visible in Figure 5, where changing the cation and anion volume factors modifies the positive and negative branches differently, see the surface free charge densities and the capacitance profiles.

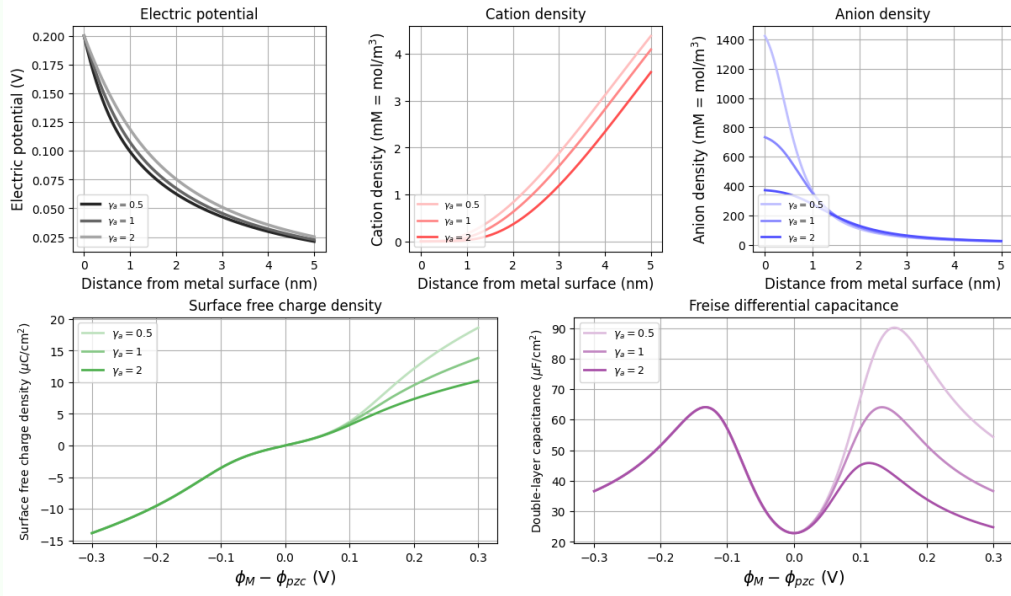


Figure 5: Freise profiles for different cation or anion volume factors. Changing γ_a mainly affects the positive branch, whereas changing γ_c mainly affects the negative branch.

3.2) Increasing γ_a near a positively charged surface

For a positively charged metal surface,

$$\Delta\phi > 0 \quad (111)$$

the potential is positive near the surface:

$$U(x) > 0 \quad \text{close to the metal} \quad (112)$$

The anions are then the counter-ions, because

$$c_-^{(F)}(x) = \frac{c_b e^{+U(x)}}{D_F(U(x))} \quad (113)$$

For $U > 0$, the anion part of the denominator contains the term

$$\gamma_a (e^U - 1) \quad (114)$$

Increasing γ_a therefore increases the denominator in the region where anions would otherwise accumulate. This reduces the near-surface anion concentration. Thus,

$$\boxed{\gamma_a \uparrow \implies c_-^{(F)}(x) \text{ near a positively charged surface decreases}}$$

Physically, a larger anion volume factor means that anions occupy more effective volume, so fewer anions can pack close to the positively charged surface. The positive-potential branch in Figure 5 shows that changing γ_a mainly affects the anion-rich side of the double layer.

3.3) Increasing γ_c near a negatively charged surface

For a negatively charged metal surface,

$$\Delta\phi < 0, \quad (115)$$

the potential is negative near the surface:

$$U(x) < 0 \quad \text{close to the metal.} \quad (116)$$

The cations are then the counter-ions, because

$$c_+^{(F)}(x) = \frac{c_b e^{-U(x)}}{D_F(U(x))}. \quad (117)$$

For $U < 0$, the cation part of the denominator contains the term

$$\gamma_c (e^{-U} - 1). \quad (118)$$

Increasing γ_c therefore increases the denominator in the region where cations would otherwise accumulate. This reduces the near-surface cation concentration. Thus,

$$\boxed{\gamma_c \uparrow \implies c_+^{(F)}(x) \text{ near a negatively charged surface decreases.}}$$

Physically, a larger cation volume factor means that cations occupy more effective volume, so fewer cations can pack close to the negatively charged surface. The negative-potential branch in Figure 5 shows that changing γ_a mainly affects the anion-rich side (right side) of the double layer capacitance and surface free charge density.

3.4) Asymmetry of the surface free charge density

The surface free charge density is defined by

$$\sigma_{\text{free}} = -\varepsilon_0 \varepsilon_S \left. \frac{d\phi}{dx} \right|_{x=0^+}. \quad (119)$$

In the symmetric Bikerman case, the steric response is the same for positive and negative potentials. Therefore, the charge-voltage curve is antisymmetric:

$$\sigma_{\text{free}}(-\Delta\phi) = -\sigma_{\text{free}}(\Delta\phi). \quad (120)$$

This symmetry is lost when

$$\gamma_c \neq \gamma_a. \quad (121)$$

For positive potentials, the response is mainly controlled by the anion volume factor γ_a . For negative potentials, the response is mainly controlled by the cation volume factor γ_c . If these two factors differ, the positive and negative branches are affected differently. Hence,

$$\boxed{\gamma_c \neq \gamma_a \implies \sigma_{\text{free}}(\Delta\phi) \text{ is asymmetric around } \Delta\phi = 0.}$$

At the potential of zero charge itself,

$$\Delta\phi = 0, \quad (122)$$

one still has

$$U(x) = 0, \quad D_F(0) = 1, \quad (123)$$

and therefore

$$\sigma_{\text{free}}(0) = 0. \quad (124)$$

The asymmetry appears in the positive and negative branches away from the PZC.

3.5) Asymmetry of the differential capacitance

The differential capacitance is defined as

$$C_{\text{dl}}^{(\text{F})} = \frac{d\sigma_{\text{free}}^{(\text{F})}}{d\Delta\phi}. \quad (125)$$

Since the surface charge density becomes asymmetric when $\gamma_c \neq \gamma_a$, its derivative with respect to $\Delta\phi$ is also generally asymmetric.

For positive potentials, the capacitance is mainly influenced by how the anion accumulation is limited by γ_a . For negative potentials, it is mainly influenced by how the cation accumulation is limited by γ_c . If

$$\gamma_a \neq \gamma_c, \quad (126)$$

then the two sides of the capacitance curve no longer have to match.

Thus,

$$\boxed{\gamma_c \neq \gamma_a \implies C_{\text{dl}}(\Delta\phi) \text{ is generally asymmetric around the PZC.}}$$

The surface charge and capacitance panels in Figure 5 show the resulting asymmetry in σ_{free} and C_{dl} .

3.6) Comparison with Valette data

The Freise model improves upon the symmetric Bikerman model by allowing the positive and negative branches to differ. This is useful because experimental capacitance curves are often not perfectly symmetric around the potential of zero charge.

Compared with the symmetric Bikerman model, Freise can therefore provide a better phenomenological fit if the experimental data show different behavior on the positive and negative sides of the PZC. In particular,

- γ_a can tune the positive-potential branch, where anions are the counter-ions,

- γ_c can tune the negative-potential branch, where cations are the counter-ions.

However, the Freise model is still a finite-size mean-field model. It does not explicitly include specific adsorption, dielectric saturation, solvent layering, ion-ion correlations, or chemical interactions with the metal surface. Therefore, even if it improves the agreement compared with symmetric Bikerman, it should not be interpreted as a complete microscopic description of the interface.

Hence,

Freise can improve asymmetric fits to Valette data,

but

additional interfacial physics may still be needed for quantitative agreement.

As shown in Figure 6, the Freise model improves upon the symmetric Bikerman model by allowing the positive and negative branches to differ. This is useful because experimental capacitance curves are often not perfectly symmetric around the potential of zero charge.

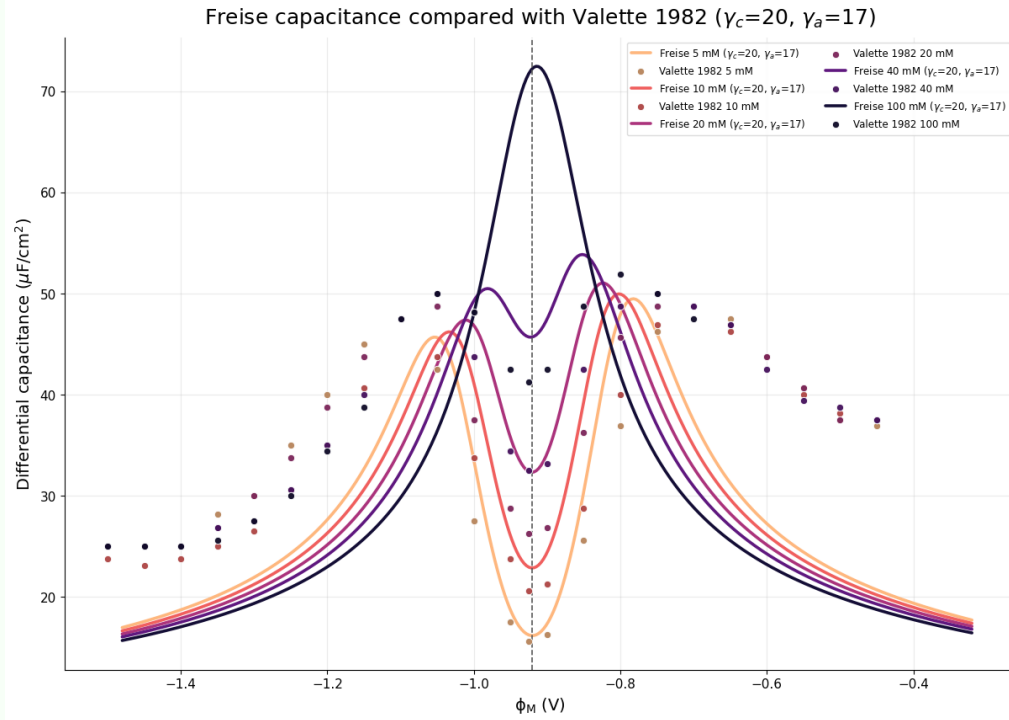


Figure 6: Comparison between the asymmetric Freise capacitance and experimental Valette data. Different cation and anion volume factors can tune the two branches separately, but remaining deviations indicate missing interfacial physics.

References

- [1] O. Schmidt, “Exercise 3: Electric double layer modeling: Gouy–chapman–stern & biker-man,” 2026. Problem sheet.